

FERGUS FALLS, MN, USA

WMO: 726572

Lat: 46.283N

Lon: 96.150W

Elev: 361

StdP: 97.07

Time Zone: -6.00 (NAC)

Period: 96-19

WBAN: 94966

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -28.6	(c) -26.2	(d) -32.4	(e) 0.2	(f) -28.5	(g) -29.0	(h) 0.3	(i) -25.4	(j) 15.5	(k) -13.6	(l) 14.0	(m) -13.0	(n) 2.9	(o) 320	(p) 0.622

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 10.9	(c) 31.3	(d) 22.3	(e) 29.1	(f) 21.2	(g) 27.8	(h) 20.5	(i) 24.3	(j) 29.0	(k) 23.2	(l) 27.5	(m) 22.1	(n) 26.5	(o) 6.1	(p) 180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
22.7	18.2	27.3	22.1	17.5	26.6	21.0	16.4	25.2	75.7	29.0	70.8	27.3	66.4	26.0	29.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12.3	11.0	9.5	-32.5	34.4	3.0	2.2	-34.6	36.0	-36.4	37.2	-38.0	38.5	-40.2	40.0
			-32.6	26.5	2.9	1.2	-34.7	27.3	-36.5	28.0	-38.1	28.7	-40.3	29.6
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	6.2	-11.7	-10.7	-1.8	6.5	13.9	19.6	22.1	20.5	16.2	8.0	-0.7	-9.2
	DBStd	13.45	8.25	7.49	7.55	6.03	5.17	3.53	3.05	3.03	4.60	5.27	6.78	7.20
	HDD10.0	2805	672	578	372	135	22	0	0	0	4	101	324	596
	HDD18.3	4764	931	812	626	357	153	27	6	12	94	322	572	854
	CDD10.0	1402	0	0	4	29	144	288	376	326	191	41	3	0
	CDD18.3	320	0	0	0	1	17	65	124	80	31	3	0	0
	CDH23.3	2541	0	0	1	10	184	516	1007	562	234	27	0	0
	CDH26.7	689	0	0	0	2	57	133	299	136	57	6	0	0
(14) Wind	WSAvg	4.3	4.4	4.3	4.6	4.8	4.7	4.1	3.5	3.4	4.0	4.5	4.6	4.2
(15) Precipitation	PrecAvg	587	19	14	27	54	68	96	85	80	62	44	24	17
	PrecMax	905	78	36	56	124	179	242	186	174	168	161	112	60
	PrecMin	384	2	0	6	1	12	18	11	6	8	3	2	1
	PrecStd	126	18	9	14	34	45	47	44	38	44	37	22	14
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	5.8	8.8	19.0	24.0	32.0	32.6	33.8	32.3	31.2	27.1	17.6	7.7
		MCWB	2.6	6.9	12.6	14.0	19.4	22.8	24.6	22.0	21.0	17.6	11.7	5.0
	2%	DB	2.5	3.9	14.9	21.2	27.8	30.0	32.0	29.8	28.0	22.4	13.9	3.0
		MCWB	1.0	2.3	9.9	12.0	18.0	21.0	23.6	22.5	20.2	14.5	9.0	1.2
	5%	DB	1.1	2.1	11.3	18.4	25.0	27.7	29.2	27.8	26.3	18.8	11.2	2.0
		MCWB	0.2	1.0	7.8	11.0	16.5	19.6	22.0	21.1	19.2	13.2	7.6	1.0
	10%	DB	-1.0	-0.3	7.5	16.2	22.4	26.3	27.8	27.0	23.8	16.2	7.8	0.1
		MCWB	-1.4	-0.8	4.9	9.5	15.3	18.6	21.2	20.4	18.3	11.2	5.1	-0.5
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	2.9	6.7	14.9	15.4	21.8	25.2	26.7	24.8	23.8	18.4	12.9	5.3
		MCDB	5.3	8.1	17.5	21.8	26.9	29.2	31.8	29.0	27.4	24.1	16.9	7.8
	2%	WB	1.2	2.7	11.0	13.3	19.7	23.0	24.8	23.6	22.2	15.7	10.0	2.1
		MCDB	2.3	4.0	14.0	18.8	25.1	27.7	29.9	27.7	26.2	21.0	13.4	3.2
	5%	WB	0.0	0.8	7.7	11.7	18.0	21.4	23.6	22.6	20.5	13.7	7.6	0.8
		MCDB	0.9	1.8	10.5	16.8	22.8	26.1	28.3	26.4	24.8	18.0	10.4	1.7
	10%	WB	-1.5	-1.2	4.8	10.1	16.6	20.0	22.3	21.4	19.0	11.6	5.4	-0.8
		MCDB	-1.1	-0.5	7.8	15.4	21.0	24.6	26.9	25.1	23.3	15.4	8.1	-0.1
(35) Mean Daily Temperature Range	5% DB	MDBR	9.7	9.9	9.4	10.9	11.5	10.6	10.9	10.7	10.9	9.7	8.7	8.5
		MCDBR	10.4	10.8	13.2	15.6	15.2	12.5	12.7	12.2	13.3	13.6	12.6	9.6
	5% WB	MCWBR	8.7	9.0	8.7	8.8	7.9	6.3	6.5	6.0	7.0	7.9	8.2	7.9
		MCDWR	10.0	10.9	12.8	13.8	12.8	11.1	11.2	10.8	11.3	12.1	11.5	8.9
(40) Clear-Sky Solar Irradiance	taub		0.282	0.291	0.303	0.350	0.370	0.380	0.381	0.382	0.351	0.313	0.284	0.264
		taud	2.317	2.316	2.429	2.360	2.372	2.384	2.376	2.372	2.447	2.534	2.515	2.417
	Ebn at Noon		814	884	930	906	896	885	879	864	864	857	822	807
		Edh at Noon	76	93	97	115	117	117	116	112	95	75	63	61
(44) All-Sky Solar Radiation	RadAvg		1.68	2.80	3.84	4.87	5.47	6.07	6.41	5.38	4.07	2.54	1.60	1.19
	RadStd		0.18	0.30	0.37	0.39	0.39	0.39	0.26	0.47	0.28	0.36	0.17	0.14

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (37 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page